MIL-T-60150A(MU)
12 August 1966
SUPERSEDING
MIL-T-60150(MU)
24 December 1964

MILITARY SPECIFICATION

TELESCOPE, STRAIGHT: 8287286

1. SCOPE

1.1 This specification covers one type of 6-power prismatic telescope which has a 12-degree field of view.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-5400	Electronic Equipment, Aircraft, General Specification For
MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel; Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	**Sampling Procedure and Tables for Inspection by Attributes**
MIL-STD-109	**Quality Assurance Terms and Definition:**

DRAWINGS

U.S. Army, Frankford Arsenal

F8287286	Telescope, Straight 8287286

FSC **1270**

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PACKAGING DATA SHEET

MIL-P-14232/P8287286 Packaging of Telescope, Straight: 8287286

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. **REQUIREMENTS**

3.1 First article testing.- Requirements for the submission of first article samples--by the contractor for testing shall be as specified in the contract.

3.2 **General specification.- The contractor shall be responsible for adherence to the-requirements of MIL-F-13926 specified below:**

(a) Dimensions and tolerances
(b) Inorganic protective surface finishes
(c) Part identification and marking
(d) Worlmanship

3.3 Fabrication. - The telescope shall be fabricated in accordance with Drawing F8287286 and drawings pertaining thereto.

3.4 **Environments 1.**

3.4.1 Storage temperatures.- The telescope shall show no evidence of physical failure such as glass breakage cement failure, or casting failure due to storage at temperatures -80° and +160° Fahrenheit (F). Subsequent to meeting this requirement, the telescope shall be capable of meeting the requirements of 3.4.3.

3.4.2 Standard temperature.- The telescope shall meet the following requirements while exposed to temperatures of 60° and 90°F.

3.4.3 Vibration.- The telescope shall meet the requirements of this specification after being subjected to vibrations which comply with curve III of MIL-E-5400.

3.5 Sealing.- The telescope shall show no evidence of leakage when subjected to an external pressure of 5 psi gage pressure.

3.6 **Cleanliness. - The telescope Shall meet the cleanliness re= quirements of MIL-0-13830 and there shall be** no more than 3 particles of dirt visible when viewed through the eyepiece. The area of each particle shall not be in excess of the area equivalent to a size 10 dig. There shall be no dirt on other surfaces which interferes with observation or adversely affects image quality.

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3.7 Image tilt.- The image of a vertical target line shall emerge vertical within one degree of arc.

3.8 Focus .- When viewing a target at 2000 meters, the diopter setting of the eyepiece shall be fixed within the limits of -0.50 and zero diopter.

3.9 Magnification. - Magnification shall be within the limits of 5.7 and 6.3 power.

3.10 Resolution.- Resolution on the optical axis shall be 6.0 seconds of arc or less.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article testing.- When the contract specifies testing of preproduction or initial production samples, the following paragraphs shall apply as applicable.

4.2.1 Preproduction test.- Preproduction test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, two preproduction samples shall be submitted to the procuring agency for approval of design and construction features. Preproduction samples shall meet all inspection provisions of 4.3, 4.4 and 4.6 as well as any additional tests that may be specified in the contract.

4.2.2 Initial production test.- Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials processes and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4 and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may 'be deemed necessary.

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4.2.3 First article disapproval.- Failure of any one of a first article sample submitted shall be cause for disapproval by the Government and subsequent action as required by the contract clauses applicable to forst article testing.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and -tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL), when provided with the technical data package (TDP). In the absence of SQAP'S the applicable Quality Assurance Provision of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I and II herein. Examination and tests for Packaging, Packing arid Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I and II shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS		REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR: AQL 1.5 Percent Defective			
101.	vibration	3.4.3	4.6.1.2
102.	Seal ing	3.5	4.6.2
103.	Cleanliness	3.6	4.6.3
104.	Image tilt	3.7	4.6.4
10.5.	Focus	3.8	4.6.5
106.	Magnification	3.9	4.6.6
107.	Resolution	3.10	4.6.7
MINOR: NONE			

NOTE : The tests in Table II shall be conducted at a temperature between 600 and 90°F.

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4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.- Three telescopes shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE 11

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
401. General specification	3.2	MIL-F-13926 - visual
402. Fabrication	3.3	Applicable Drawing - visual
403. Temperature (storage)	3.4.1	4.6.2.1

4.4.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, Supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or **SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the technical data package (TDP).**

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

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4.6 Test methods, and procedures.

4.6.1 Environmental.

4.6.1.1 Storage temperature.- The telescope, while in a non-operating condition shall be subjected to the Temperature Cycling Test, Procedure 11 of MIL-F-13926 using Conditions A and H. Upon stabilization at ambient room temperature, the telescope shall meet the requirements specified in 3.4.1.

4.6.1.2 Vibration test.- With the telescope mounted in the normal operating position, the table shall be vibrated along a horizontal direction with the frequency varying between 5 and 500 cps. For frequencies between 5 and 20 cps, "the double amplitude shall be 0.1 inch. For frequencies between 20 and 33 cps, the amplitude shall be varied so as to maintain a 2 G acceleration. For frequencies between 33 and 52 cps, the double amplitude shall be 0.036 inch. For frequencies between 52 and 500 cps, the amplitude Shall be varied so as to maintain a 5 G acceleration. The rate of frequency change shall be logarithmic and shall be such that a complete cycle (5-500-S) will consume approximately 15 minutes. The test shall continue for four 15-minute cycles for a total of 1 hour. (When provisions for logarithmic cycling is not available, a linear rate of frequency change may be used.) The cycling test shall be repeated with the axis of vibration changed 90 degrees in the horizontal direction and then with the axis of vibration change to vertical. At the completion of each direction of vibration, the telescope shall be closely inspected for any evidence of mechanical failure, loose or broken parts and defective optics. Subsequently, the telescope shall meet the requirements of 3.4.3.

4.6.2 Sealing.- The telescope shall be externally pressurized to the psig value specified in 3.5, using commercial test equipment capable of obtaining the specified pressure within 1/4 psig. The external pressure shall be applied for a minimum of 5 minutes and shall show no decrease in pressure while the telescope is 'tinder test to determine conformance with 3.5.

4.6.3 .Cleanliness.- The optical surfaces of the completed telescope shall be inspected for cleanliness in accordance with the requirements of 3.6 and MIL-O-13830. Any dirt particles which interfere with the visibility of the telescope or exceeds the dig size specified in 3.6 shall "be cause for rejection. The mean diameter of a size 10 dig is equivalent 'to O.10 millimeter.

4.6.4 Image tilt.- Using a collimating telescope of approximately 3 power, align the vertical reticle cross line with a plumb line. .In-'troduce 'the telescope under test into the "line of sight between the target and the collimating telescope. Any deviation of the image of the plumb line with relation to the vertical line of the collimating telescope shall not exceed "the tolerance of 3.7.

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4.6.5 Focus.- Using a calibrated dioptometer of at least 3 power, focus the eyepiece on the reticle, then position the dioptometer on the optical axis of the telescope at its eyepiece to view a 2000-meter target. Adjust the diopter setting to obtain the sharpest image of the target, The diopter setting thus established shall be within the limits specified in 3.8.

4.6.6 Magnification.- Position a 20mm diaphragm over the objective lens and place a light source *near* the diaphragm. Position a dynameter in front of the eyepiece, and adjust for sharp focus of the 20mm diaphragm. Count the number of lines on the dynameter reticle that it takes to span The diaphragm at its largest diameter and divide this figure into the diameter of the diaphragm. The result shall be within the tolerance as specified in 3.9.

4.6.7 Resolution.- This test shall be performed in accordance with the resolution test in MIL-0-13830, and the telescope shall meet the resolution value specified in 3.10.

5. PREPARATION FOR DELIVERY

5.1 Packaging, Packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P8287286. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The telescope: 8287286 is primarily intended for use as a component of the Antioscillation Sighting System XM58 used in a UH-lB Helicopter for observing the flight of a guided missile.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet number (see 5.1).

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Project No. 1270-A040Y

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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